

ORGANIZATION:

An organization is a group of people working together in a formal way.

Explanation: The work that must be done is shared between these people and there are rules about who does what.

How the work is shared and how tasks and people are grouped together – the structure of the organization – will vary very much from organization to organization. It is surprising, however, that organizational structures have much more in common than might be expected. Organizational theory is the study of how organizations are structured and how they work.

ORGANIZATIONAL MODEL:

1. The bureaucratic model

All tasks are split up into specialized jobs, in which jobholders become experts; management can thereby hold the jobholders responsible for the effective performance of their duties. Precise rules govern the performance of each task. This means that there should be no variation in the way tasks are carried out and therefore no problems with the coordination of different tasks.

Everyone (and hence each unit) in the organization is accountable to one and only one manager.

To ensure that personalities and personal relationships do not interfere with the organization's performance,

employees must relate both to other employees and clients in an impersonal and formal manner.

Recruitment is based on qualifications and employees are protected against arbitrary dismissal. Promotion is based on seniority and achievement. Lifetime employment is envisaged. In a modified form, this model still describes the organizational structures found in most large and many smaller organizations.

The vertical world of hierarchical bureaucracy:

Roles, Rules, Plans, Reports
Boss
/ | \
Individuals

2. The organic model (best-known alternative model)

"An organization will be effective to the extent that its structure is such as to ensure a maximum probability that in all interactions and in relationships within the organization, each member, in the light of his background, values, desires, and expectations, will view the experience as supportive and one which builds a sense of personal worth and importance."

This view underlies the organizational structure of most small professional companies – software houses, advertising agencies, and even solicitors' and GPs' practices

It is also common in academic institutions, both schools and universities. The

view is not necessarily consciously articulated – nor is this view and the adoption of the structures it suggests sufficient to achieve effectiveness!

Proponents of the bureaucratic model claim that it is universally applicable.

Proponents of the organic model make similar claims.

Common sense is that those who believe each has its appropriate place should be of the contingency school of organizational design.

3. Matrix management

In a bureaucratic model every individual and every unit in the organization is responsible to only one manager. This is not realistic in the context of project-based, high-technology companies. MM believes that individuals may be responsible to more than one manager and requires rules that will enable possible conflicts to be resolved.

For instance, a specialist in high-speed communications working for a systems integrator may well find themselves working on two or three projects simultaneously, as well as having a more general responsibility for maintaining the company's expertise in the area. In the past 30 years or so the idea of matrix management has become fashionable as a way of addressing such situations. However, the results are not encouraging.

STRUCTURING PRINCIPLES:

1. Structure by function

There are the activities that are the primary purpose of the organization. These activities are known as operations.

All organizations services or pay their bills and pay their employees. They need to ensure that the buildings they use are cleaned regularly. If they charge for their services, they may need to send out bills and ensure that these are paid.

They will probably need to hire new employees from time to time. These activities are generally known as administration.

Many organizations will need to publicize their services or products and try to persuade people to use or buy them. In the business world, these activities are usually known as sales and marketing.

Finally, many organizations need to be continually developing new products or services or developing new ways to deliver them. These activities are known as research and development.

C o m p a n y

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Operations SnM Admin RnD

Advantages

Permits centralized control of strategic results

Promotes in-depth functional expertise

Enhances operating efficiency where tasks are routine and repetitive.

Disadvantages

Poses problems of functional coordination

Can lead to interfunctional rivalry, conflict, and empire-building

May promote overspecialization/ narrow viewpoints

2. Structure by geography

In many cases it makes sense to group activities on a geographical basis.

Multinational companies, that is, companies that operate in several different countries, are usually forced to have some geographical elements in their structure.

The subsidiaries are subject to the laws of the countries in which they are registered, in particular, the laws regarding employment, accounting, and taxation.

Within a single country, geographical factors have become less important as a result of the development of modern communications and, as a result, geographical structures have been replaced by structures based on other factors.

Advantages

Allows tailoring of strategy to needs of each geographic market

Improves coordination within the market

Takes advantage of economies of local operations

Disadvantages

Greater difficulty in maintaining consistent company image across areas
Can result in duplication of staff services at headquarters and district levels, making a relative-cost disadvantage

3. Product line structure

A product line structure is a structure that is based around the different types of product that an organization produces. For example, where a motor vehicle manufacturer organizes around types of vehicle.

Companies that produce and market a substantial piece of software for corporate customers – a multi-user accounting package

For example – often organize themselves into three main operational divisions:

development and maintenance of the software, consultancy, and training.

This should be regarded as a product line structure since the three types of activity, providing software, giving advice to companies on how to use it, and providing training for customer staff, can be considered to be different services that the company provides and they are typically provided by different teams of people.

4. Structure by market sector

means structure based on the different market sectors to which its customers or prospective customers belong.

From the sales and marketing point of view it has the great advantage that each division can readily identify its potential customers, and its staff, both sales and technical, are likely to be familiar with customers' problems and to speak a language that the customer understands.

Example : Different branches of BATA, Automobiles

Disadvantages: There are two dangers with this approach.

There is the risk one division may be unaware of technological expertise that exists in another division. This may lead to inefficient use of resources through unnecessarily hiring additional specialists or employing consultants, or, worse, to failing to learn from mistakes that have been made by other parts of the company.

The second danger with a structure based on the market sector is that, by continuing to concentrate on its traditional areas even when these markets are becoming saturated, the company will miss new opportunities and will stagnate.

5. Structure by technology

A technology-based structure was once a favorite model for software companies.

A company might have divisions specializing in artificial intelligence, communications, web-based systems, databases, and real-time systems.

There are several problems with type of structure: it usually requires several different technologies to meet a customer's needs; there are many applications that cannot be said to require specific technologies; there are many competent software engineers whose expertise runs across

a number of technologies; it is difficult, if not impossible, for sales and marketing staff to predict which potential clients will need which technology. It is particularly serious and companies that are primarily structured by technology have serious problems finding their clients. They are not sufficiently 'customer-focused' – they concentrate on selling the technologies that they have rather than finding out what the customer needs.

6. Operational structure

The actual operations of a company may be organized on a project basis or on a production basis, although the line separating the two may be vague.

Project-based activity is not restricted to operations. Most research and development is organized on a project basis and such administrative activities as introducing a new accounting system or transferring a company's head office are also to be regarded as projects, in that they last, or a fixed length of time, after which they should be complete.

Projects last a comparatively long time but the team carrying out the work only stays together for the length of the project. Production activities are comparatively short, but the team carrying them out stays in existence indefinitely.

DEPTH OF STRUCTURE:

The depth of an organizational structure is the number of layers in the

structure – or, more precisely, the maximum number of layers, since not all parts of the structure will have the same number of layers.

Organizational structures are often described as flat or, in contrast, deep or tall, according to whether the depth is small or large. For a given number of people, the depth of the structure will depend on the number of people reporting directly to each manager; this is sometimes known as the manager's span of control.

CENTRALIZATION:

Organizations may be centralized or decentralized. In a centralized company, as much power as possible is kept at the top of the company, with delegation only when essential.

In a decentralized company, as much power and control as possible is delegated to the lowest level.

SETTING UP A

STRUCTURE IN PRACTICE:

In most cases, an organization of any size will have a structure that includes elements of several of the different types of structure. A combination of different structures would help when adopted.

NEGATIVE RIGHTS:

Negative rights are those rights which require restraint by others.

Example: My right not to be killed, robbed, poisoned, harmed and so on. It requires others to refrain from killing, robbing, poisoning me.

However negative rights does not require that others take action to keep me alive, putting locks and guards to save me from being robbed.

POSITIVE RIGHTS:

Positive rights are those which implies others have a duty towards the right holder. Example: If I have a right to live then not only others have to refrain from killing me but also they need to feed me, keep me safe etc.

LEGAL RIGHTS:

Legal rights are rights created and applied by the law. The law enforces a statement or code of conduct, a deed to be done or rule to follow and fixes a punishment against violating the law.

MORAL RIGHTS:

Moral , natural or human rights are independent from the law.

A civilized society's first line of defense is not the law, police and courts but customs, traditions and moral values. Behavioral norms, mostly transmitted by example, word of mouth and religious teachings, represent a body of wisdom distilled over the ages through experience and trial and error.

Moral status leads to let you decide on the base of good or bad consciousness, religious training, manners etc. Although it does not involve penalty in violating such rights, a feeling of guilt is always there to follow.

Legal status enforces your deeds on the grounds of defined law and has a

penalty or punishment for those violating such rights or laws.

Case 1 (Polluting the environment)

Moral status: Don't throw waste papers on the ground it will pollute the environment

Legal status: Making the case legal how????

Case 2 (Oldies get a seat in a loaded bus)

Moral Status: Youngers should give the oldies a seat ; good manners

Legal Status: Label some seats for the oldies and fine those youngsters who don't give away the seats for the oldies

Case 3 (Password privacy in an organization)

Moral status: No one is supposed to share their password else their data can be tampered, stolen etc..

Legal status: If any one leaks his password he should pay a fine and be suspended.

Case 4 (An institute giving degrees to those who are not eligible vs giving degrees which are not recognized)

Moral status: No one is supposed to give degrees unless the requirements are fulfilled. Low standards of required skill are against moral values.

Legal status: If someone provides degrees which are not recognized by any higher commission or whose charter is not valid.

OCCUPATION:

A person involved in a job and earning money is having that job as an occupation

Example: A doctor teaching in a school, college or university has an occupation of teaching, carpenter, driver

PROFESSION VS

OCCUPATION:

A profession needs extensive training and specialized knowledge. On the other hand, an occupation does not need any extensive training.

A profession can be called an occupation when a person is paid for his particular skills, and his deep knowledge.

Persons engaged in an occupation are not paid for their knowledge, but only for what they produce.

Unlike a person engaged in an occupation, a professional has to undergo higher education.

A profession tends to be autonomous, whereas, for an occupation, no one has autonomous power; he is supervised by another person.

Unlike occupation, a profession demands that the responsibility lies with the individual.

A profession is guided through certain ethical codes, and regulated by certain statute

PROFESSION:

Mastery of an Esoteric Body of knowledge: usually acquired through higher education

A member of the profession needs this body of knowledge in order to practice

A related characteristic of professions is that they often embrace a division between researchers and practitioners

Autonomy: Professionals generally have a great deal of autonomy, justified by having an esoteric knowledge Professions typically have autonomy both at the collective level as well as in individual practice

Profession sets its own standards rather than taking orders or suggestion from outsiders

Formal Organization:

Generally there exists a professional organization that:

Controls admission to the profession and sets standards

Involved in licensing & expelling its individual members

Code of ethics

There is a code of ethics & members of professions must adhere to the code no matter what their employment context (take an oath)

Social function: Professions generally fulfill an important social function and for the welfare of the society

Medicine (promoting health) lawyers not alone but play an important role in justice.

PROFESSIONAL BODIES:

A profession is typically organized into one or more professional bodies. What is a professional body?

A professional association (also called a professional body, professional organization, or professional society) is usually a nonprofit organization seeking to further a particular profession, the interests of individuals engaged in that

profession and the public interest. OR

A professional body is a group of people in a learned occupation who are entrusted with maintaining control or oversight of the practice of the occupation

A professional body usually starts by a group of people coming together because of a shared interest in a particular type of activity.

Functions of Professional bodies:

As the professional body matures, it is likely to develop a range of functions, of which the following are the most important:

establishing a code of conduct to regulate the way members of the body behave in their professional lives and a disciplinary procedure to discipline members who breach(fail to follow or conduct) this code;

establishing mechanisms for disseminating(spreading) knowledge of good practice and new developments to its members, typically through publications and conferences but increasingly also through the use of the worldwide web; setting standards of education and experience that must be met by people wishing to become members of the body;

advising government and regulatory bodies about matters within its area of expertise.

RESERVATION OF TITLE AND FUNCTION:

The use of the name of the profession may be restricted

to those people who are appropriately qualified.A restriction of this sort is called reservation of title.

The law may state that certain activities are restricted to people with appropriate qualifications.

This is called reservation of function.

Is Computing A Profession

The field of computing is so diverse and complex

The range of those who are called computer professionals is extremely broad

Nothing in common except they involve the use of computer

At first glance, it seems that computing possesses all of these characteristics, though in complex ways

Are Computer

Professionals

“Professionals”

Computer professionals have mastered an esoteric body of knowledge

Computer professionals have varying degrees of autonomy depending on where they work

There is no single organization governing the profession of computing (ACM, IEEE)

Computing is an activity which supports social institutions which in turn are aimed at fulfilling a variety of social functions

Computing does not appear to be a 'profession', the way law and medicine are professions, but it is more of a 'profession' than many other occupations.

SOFTWARE ENGINEERING

Software engineering appears to be one area of computing that is emerging as a distinct profession within the field of computing.

This trend seems to have been created by the initiative of individuals in the field concerned about the quality and safety of the software being produced and sold. Developing software engineering as a distinct field involves several of the activities.

For example, it means identifying a unique body of knowledge that a person must possess to be a competent software engineer. It means developing educational requirements (curriculum) such that the person who meets the requirements is more likely to produce a quality, safe software than someone without training. It means developing mechanisms for licensing of members.

This will include identifying or creating the proper organization for issuing licenses and identifying requirements for obtaining a license, such as passing an exam or acquiring a certain number of years of experience.

Another requirement for professionalization is a code of ethics.

The development of software and information systems in general is now usually regarded as a branch of engineering and the people who practice

it are considered to be engineers.

Traditional engineers design and build a wide variety of objects – dams, bridges, aeroplanes, cars, radio and television transmitters and receivers, computers, plants to make fertilizer or plastics, and so on.

There are two constraints that apply to all such activities and which can be regarded as characteristic of engineering:

First engineering involves designing and building things that must work properly, that is, must meet a set of predetermined requirements concerning their functionality, their performance, and their reliability;

Second the process of designing and building the object must be completed within specified constraints of time and budget.

THE STATUS OF ENGINEERS

it is illegal to call yourself an engineer in a given state unless you are registered with the State Engineers Registration Board; it is illegal for a company to use the word 'engineering' in its name unless it employs at least one registered engineer; academic programmes including the term engineering in their title must be taught mostly by registered engineers; it is illegal to carry out engineering work except under the

supervision of a registered engineer.

Washington Accord

The Washington Accord is an international accreditation agreement for professional engineering academic degrees, between the bodies responsible for accreditation in its signatory countries. There are three agreements covering mutual recognition in respect of qualifications in engineering:

The Washington Accord signed in 1989 was the first - it recognizes substantial equivalence in the accreditation of qualifications in professional engineering, normally of four years duration.

The Sydney Accord commenced in 2001 and recognizes substantial equivalence in the accreditation of qualifications in engineering technology, normally of three years duration.

The Dublin Accord is an agreement for substantial equivalence in the accreditation of tertiary qualifications in technician engineering, normally of two years duration. It commenced in 2002.

Professional Bodies in Computing

The development of Professional Bodies in Computing

The Institute of Electrical and Electronic Engineers (IEEE): A professional engineering society basically USA based but with members and

activities spread worldwide. This was the IEEE Computer Society (IEEE-CS). It has over 100,000 members.

Association for Computing Machinery (ACM):

USA based but have members and activities in many countries. It has over 75,000 members

British Computer Society (BCS):UK based society. It became a qualification awarding society. Then it named itself as BCS- The Chartered Institute of IT.

Institution of Electrical Engineers (IEE): It has over 130,000 members.

Italian Association for Informatics and Automatic Computing

The Computer Society of India

Australian Computer Society the Singapore Computer Society and the Irish Computer Society

German Informatics Society The Computer Society of Sri Lanka

The Computer Society of Maritius

Code of Conduct

All the mentioned professional bodies have set a code of conduct that their members need to obey.

The BCS's Code of Conduct:

The Code is divided into the following sections:

1.The Public Interest

Regard public health, privacy, security and wellbeing of public

Regard rights of third party

Conduct your professional activities without personal biasness

Promote equal access to the benefits of IT

2.Duty to the Relevant Authority

Avoid conflicts of interest.

Avoid misrepresentation

Don't pass on confidential information without permission.

3.Duty to the Profession

Accepting personal duty and avoid actions which can harm the image of the profession

Seek to improve professional standards through participation in their development, use and enforcement.

Encourage and support fellow members in their professional development

4.Professional Competence and Integrity

Only take a job or offer a service that is within your professional competence.

Do not claim any level of competence that you do not possess.

Get up to date knowledge in your relevant field.

Respect and value alternative viewpoints and, seek, accept and offer honest criticisms of work

Avoid harming others by false devilish or negligent actions

Reject bribery

BCS EDUCATION:

The BCS promotes education in a number of ways:

Professional examinations, approval to suitable organizations that provide courses to prepare students for them.

It accredits degree programs offered by institutions of higher education.

It sets the syllabus and accredits training organizations to provide the associated short courses.

1.The professional examinations

Purpose: to recognize skills and lead to qualify for membership

Stages: the Certificate, the Diploma, and the Professional Graduate Diploma.

2.Accreditation & exemption

Accreditation means that a course has fulfilled the educational requirement

A course may qualify for full or partial exemption from the Higher Education

Qualifications (HEQ), and students graduating from such courses can attain

Professional Membership after a shortened period of relevant experience and training

3.Short courses

BCS offers a substantial range of qualifications achievable through short courses.

The courses are intended as training courses for staff working in the industry.

BCS—CONTINUING PROFESSIONAL DEVELOPMENT

"The systematic maintenance, improvement and broadening of knowledge and skill and the development of personal qualities necessary for the execution of professional and

technical duties throughout the individual's working life."

BCS—CPD—activities

CPD services to individual members: Computer Bulletin ,keeping up-to-date info of latest computer areas and development
Career development and CPD services to the industry: Resolving IT staff issues of placement in the industry and training

BCS—THE ADVANCEMENT OF KNOWLEDGE

One of the first actions of the BCS when it was formed was to establish The Computer Journal. Currently, six issues a year are published.

BCS—MEMBERSHIP GRADES

1. The standard grades:

- a.Affiliate: open to anyone with an interest
- b.Companion: members of other professional bodies with at least five years' work experience related to information systems
- c.Student: open to those following an approved course of study leading to a qualification recognized for admission to Associate or Professional membership of the Society.
- d.Associate: Associate membership is open to anyone who has any of the following:
 - ▶ more than one but less than five years of IT work experience;
 - ▶ a Higher National Certificate in an IT-related field;
 - ▶ a non-accredited degree with a significant IT content.

2. The professional grades

a.Membership: In order to qualify for membership in the professional grades, you must have one of the following:

- ▶ the BCS Professional Graduate Diploma including the project;
- ▶ an honours degree that gives you full exemption from the BCS examinations;
- ▶ five years of relevant professional IT work experience;
- ▶ some combination of academic qualifications and professional IT work experience that is judged to be equivalent to one of the above.

b.Fellow: Fellow is the most senior professional grade. It is open to applicants who can demonstrate a minimum of five years IT practitioner experience, senior post and authority in that field.

3. The chartered professional grades

Member or Fellow holding the BCS Professional Graduate Diploma including the project or an honors degree that gives you full exemption from the BCS examinations and you must have five years of relevant professional IT work experience.

▶ The Professional Advice Register

▶ It maintains a register recording their members in relevant areas

A.Chartered Information Technology Professional

B.Chartered Engineer

c.Chartered Scientist